

Berrien County Medical Control Authority
Procedures
BLOOD ADMINISTRATION

Initial Date: 6/15/2026

Section 7-31

BLOOD PRODUCT ADMINISTRATION



This protocol is for Paramedic use only

Purpose: To define the indications and process for transfusion of blood products within the Berrien County Medical Control Authority.

Note:

1. Only BCMCA approved transfusionists may administer blood or blood products.
2. All EMS providers must know the indications and contraindications of blood administration and when to request them.

Definitions:

1. Blood Products – Therapeutic substances derived from human blood
 - a. Low-titer whole blood (LTWB) – Whole blood that has low levels of anti-A and anti-B antibodies.
 - b. Packed Red Blood Cells (PRBC) – Concentrated preparation of red blood cells that is obtained from whole blood by removing the plasma
 - c. Liquid Plasma (LP) – blood product made from the liquid portion of whole blood.
2. Transfusionist- BCMCA credentialed paramedic who has completed prehospital transfusionist training.

Indications:

Suspected hemorrhagic shock from trauma or medical cause:

Adults:

- Systolic blood pressure (SBP) less than 90 mmHg or weak or absent radial pulse, **or**
- Heart Rate (HR) greater than 100 bpm unresponsive to initial interventions, **or**
- Shock Index (HR/SBP) greater than 1.0

Pediatrics:

- Systolic BP less than $70 + (\text{Age in years} \times 2)$
- Heart Rate: 1 year: greater than 190, 2-10 years: greater than 140

AND

One or more of the following physiologic criteria indicative of hypoperfusion:

- Altered mental status (not believed to be due to intoxication or head injury)
- Pale, cool, clammy skin; pale mucosa
- Delayed capillary refill (> 2 seconds)
- Tachypnea
- $\text{ETCQ} < 25$ mmHg

Exclusion Criteria:

- Isolated fall from standing injury mechanism with no active bleeding
- Suspected cervical cord injury with motor deficit and no active bleeding
- Traumatic arrest with > 5 minutes of ongoing CPR and no reversible cause of cardiac arrest (extremity hemorrhage or suspected tension pneumothorax)
- Brain matter exposed or isolated penetrating brain injury (GSW)
- Objection to transfusion voiced by patient or medical power of attorney

Berrien County Medical Control Authority
Procedures
BLOOD ADMINISTRATION

Initial Date: 5/14/2026

Section 7-31

- Isolated drowning or hanging victims
- Isolated burns > estimated 20% total body surface area

Treatment:

1. Follow **General Pre-hospital Care- Treatment Protocol**.
2. Control bleeding according to **Bleeding Control (BCON)- Treatment Protocol** when applicable.
3. Administer TXA per the **Hemorrhagic Shock- Treatment Protocol**.
4. Transport according to **Adult and Pediatric Trauma Triage- Treatment Protocol** and MCA transport Protocol.
5. No intervention should delay transport
6. If blood products are indicated, contact Medical Control for orders.
7. Obtain vascular access
 - a. 2 large bore (≥ 20 g) IVs recommended. One connected to blood tubing. One connected to IV 0.9% NS TKO.
 - i. Second IV can be started after blood products are initiated.
 - b. IO access may be utilized if two attempts at IV access are unsuccessful
 - i. Humeral head is the preferred site followed by tibial.
8. Two BCMCA active personnel with a license level of EMT or higher (one must be a paramedic with BCMCA-approved blood administration training) must confirm the following on the blood products:
 - a. Check the expiration date
 - b. Inspect for any signs of leakage or damage to the container
 - c. Verify blood type
 - d. Verify Rh factor
 - e. Check temperature on the data logger
 - f. Document verbal or implied consent
 - g. If any discrepancy exists in the above checks. **DO NOT PROCEED WITH THE TRANSFUSION**
9. Once blood products are verified for transfusion, administer:

Adult (15 and above): 1-unit LTWB

Pediatric (14 and below): 10mL/kg LTWB

If LTWB is not available, administer in sequence

Adult (15 and above)

First- 1-unit Liquid plasma

Then- 1-unit PRBCs

Pediatric (14 and below)

First- 10 mL/kg Liquid plasma

Then- 10 mL/kg PRBCs

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Procedures
BLOOD ADMINISTRATION

Initial Date: 5/14/2026

Section 7-31

Infusion Procedure and Documentation:

1. Blood products must always be stored in an BCMCA approved blood products cooler.
2. Blood products should be infused through an BCMCA approved fluid warmer.
3. Blood products will be connected to approved blood tubing.
4. **No other medications can be infused in the IV line with the blood product.**
5. One set of vitals (HR, BP, Temperature, SPO2) must be documented prior to administration.
6. Vitals will be documented on the approved transfusion form at the following intervals
 - a. 5 minutes after initiation of transfusion
 - b. After first set of vitals, subsequent vitals every 10 minutes until blood is transfused.
 - c. Every 10 minutes post transfusion completion
7. In instances where a second unit is administered:
 - a. The process of vitals will be repeated.
 - b. Administer 1 Gram of Calcium Chloride via IV.
8. One copy of the transfusion form will be left with the patient in the emergency department, one copy will be uploaded to the EMS PCR, and one copy will be returned to the blood bank.

Special Considerations:

1. Monitor for transfusion reaction; if present, **STOP THE TRANSFUSION**, change all lines and utilize normal saline.
2. Bag blood products for return to the blood bank. **Do not dispose of the blood product and tubing: it will need to be returned to the blood bank in a biohazard bag.**
3. If necessary, treat a transfusion reaction using **Anaphylaxis/Allergic Reaction-Treatment Protocol.**

Agency Requirements:

1. Agencies must submit their blood procurement and exchange/replacement process to the BCMCA prior to initiation of their program for approval. Any changes to that process must be submitted for approval.
2. A roster of all paramedics completing the BCMCA-approved blood administration training must be submitted to the BCMCA prior to those individuals providing that intervention.
3. All blood product administrations must be submitted to the BCMCA within 72 hours for quality review.